

DSPR 136

Cariology Exam Mastery Guide

LEARNING OBJECTIVE Official objective or exam skeleton material.	PROFESSOR SIGNAL Professor-emphasized priority: focus here.	EXAM TRAP Common wrong turn or distractor logic.	VISUAL PATHWAY Arrow map for process memory.
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Study Path [Back to TOC](#)

- Start with Learning Objectives: Exam-Ready Answers. It is the exam skeleton.
- Use the module briefs to repair weak spots, especially risk assessment, fluoride, saliva, clinical diagnosis, and non-restorative management.
- Use the comparison tables for fast discrimination questions.
- Use Professor Signal items as professor-emphasized priority signals, not as verbatim quotations.

Course Architecture and Exam Blueprint [Back to TOC](#)

Syllabus topic / chapter	Core idea	Must-know mechanisms	Must-recognize clinical patterns	Common traps
Introduction and Pathophysiology	Caries is a chronic biofilm-mediated disease of mineralized tooth tissue driven by microorganisms acting on fermentable carbohydrate over time.	Host/tooth + saliva, biofilm bacteria, fermentable carbohydrate, time; pH drop; demineralization vs remineralization.	White spot as early demineralization; cavitation as later tissue breakdown; root/coronal/location classifications.	Treating caries as just a hole instead of a dynamic disease process; forgetting time and saliva/protective factors.
Clinical Appearance and Diagnosis	Diagnosis asks: lesion or not, active or arrested, cavitated or intact, observe/prevent/fill.	Visual-tactile exam, clean/dry surfaces, radiographs, reliability/validity/sensitivity/specificity/PPV/NPV.	Chalky white spots, shiny arrested lesions, root caries near CEJ/root surface, recurrent caries around restorations.	Using a sharp explorer aggressively; confusing white spot reversibility with arrested lesion reversal.
Dentinal Caries and Pulpal Involvement	As caries approaches pulp, bacterial by-products and dentinal tubule anatomy drive inflammation and pain before bacteria necessarily invade pulp.	Tubule density increases near pulp; inflammatory mediators; tissue pressure in rigid dentin; A-delta vs C fibers; reversible vs irreversible pulpitis vs necrosis.	Sharp cold pain vs lingering aching pain; hyperplastic pulpitis; calcific metamorphosis; internal resorption.	Thinking bacteria must physically invade before pulp is affected; ignoring low-compliance pulp space.
Distribution and Determinants	Distribution is measured at population, tooth, surface, and subgroup levels; risk concentrates by age, SES, race/ethnicity, saliva, fluoride, diet, and plaque.	Prevalence/incidence vs DMFT/DMFS severity; total caries vs untreated caries; DMD framework.	Pit-and-fissure proportion increased; smooth-surface proportion declined; untreated caries concentrated in vulnerable groups.	Calling DMFT prevalence; memorizing raw prevalence numbers when trends and distinctions matter more.
Fluorides: Systemic and Topical	Fluoride works mainly through topical plaque/saliva/tooth-surface chemistry, with systemic programs as population-level exposure.	Remineralization, reduced demineralization, smooth-surface protection, plaque fluoride release during pH drops.	Water fluoridation, toothpaste, rinses, gels, varnish, SDF , fluorosis , osteosclerosis.	Treating fluoride as only systemic; forgetting SDF black staining and allergy/ulceration cautions.
Diet and Dental Caries	Frequency, stickiness, fermentability, and between-meal exposure matter more than sugar as a simple yes/no category.	Sucrose/sticky carbohydrate exposure, plaque pH, Vipeholm conclusions, xylitol effects.	Sticky between-meal sugar as high-risk; hereditary fructose intolerance lower caries; xylitol as sugar substitute with limits.	Confusing diet with nutrition; assuming natural sugar absence means caries cannot occur.
Saliva and Caries Activity Tests	Saliva is a core protective system: clearance, buffering, minerals, antimicrobial proteins, and remineralization support.	Flow rate, buffering, calcium/phosphate, mucins, staterin, IgA, xerostomia causes, flow measurement.	Radiation caries, Sjogren syndrome, medication-induced dry mouth, low flow, high-risk dry-mouth patients.	Forgetting dry-mouth complaint triggers salivary flow testing; mixing stimulated and unstimulated rates.

Syllabus topic / chapter	Core idea	Must-know mechanisms	Must-recognize clinical patterns	Common traps
Caries Risk Assessment	Risk is the balance between risk/pathologic factors and protective factors; treatment planning follows risk level.	History + clinical exam + radiographs + CRA form + saliva tests if indicated; 0/1/10 point structure; 3-year lesion/restoration/missing tooth window.	Low/moderate/high/extreme risk; dry mouth; chemo/radiation; new lesions/restorations; salivary flow <0.7 ml/min .	Deciding risk from one test; missing the 36-month window; skipping recall/bitewing intervals.
Preventive Products and Case Management	Product recommendations escalate with risk; non-restorative treatment depends on tooth type, surface, and cavitation.	Fluoride products, SDF , varnish, sealants, resin infiltration, xylitol, antimicrobials, buffers, xerostomia products.	Non-cavitated occlusal lesions -> sealant/varnish; cavitated lesions -> SDF ; root lesions -> 5000 ppm fluoride/ SDF .	Memorizing brand names instead of indications; recommending restoration when non-restorative management is tested.

Learning Objectives: Exam-Ready Answers [Back to TOC](#)

LEARNING OBJECTIVE	
	Start here. This table answers the course objectives in exam-ready language before the detailed module review.

Syllabus objective	Exam-ready answer	Must-memorize details	Common wrong-answer trap	Chapter
Understand how dental caries develops	Caries develops when cariogenic biofilm bacteria metabolize fermentable carbohydrate over time, producing organic acids that lower plaque pH and dissolve tooth minerals. If demineralization exceeds remineralization repeatedly, an intact subsurface lesion can progress to cavitation.	Host/tooth + saliva, plaque/bacteria, diet/substrate, time; organic acids -> demineralization -> proteolytic breakdown/cavitation.	Leaving out time or saliva/protective factors.	1
Be knowledgeable of caries pathophysiology	Caries is a dynamic imbalance between demineralization and remineralization. pH fluctuations below the critical level favor mineral loss; saliva, calcium/phosphate, fluoride, and plaque control favor repair. Disease expression depends on risk/protective balance.	Critical pH for enamel: 5.5 ; demin/remin are continuous; caries balance model.	Treating caries as a one-way process once a white spot appears.	1
Identify clinical appearance of caries	Early lesions may appear chalky/hyper-white with intact surface; progression can show staining, surface breakdown, cavitation, root-surface soft discolored lesions, or recurrent lesions around restorations.	White spot = early demineralization; arrested lesion = shiny/lustrous, brown stain may occur but is not required; root caries = discrete, well-defined, discolored soft area on cementum/root.	Assuming brown stain always means active decay.	2
Be familiar with distribution of dental caries	Distribution can be described by population group, tooth/surface, treated vs untreated disease, total caries experience, and severity indices. Caries increases with age; high-risk groups include lower SES and certain racial/ethnic groups; pit-and-fissure disease has become proportionally more important.	Total caries = treated + untreated; untreated caries = not yet treated; DMFT/DMFS = severity/intensity, not prevalence.	Calling DMFT a prevalence measure.	4
Know risk factors and risk indicators of caries	Risk increases with visible lesions, recent caries/restorations, teeth missing due to caries, frequent between-meal sugar exposure, low fluoride exposure, visible plaque, exposed roots, appliances, special health care needs, medications reducing saliva, dry mouth, chemo/radiation, eating disorders, and drug/alcohol abuse. Protective factors include saliva, fluoride, oral hygiene, restricted	CRA uses low/moderate/high categories with 0/1/10-point logic; 36-month history matters.	Deciding risk from one test or one risk factor alone.	8

Syllabus objective	Exam-ready answer	Must-memorize details	Common wrong-answer trap	Chapter
	carbohydrate exposure, sugar substitutes, dental home, and regular care.			
Understand the role of fluorides in caries prevention	Fluoride reduces caries by strengthening the remineralization side of the balance and reducing mineral loss during acidic challenges. It acts mainly topically through saliva/plaque/tooth-surface exposure, though systemic fluoridation provides population-level delivery.	Systemic: water, salt, milk, supplements. Topical: toothpaste, rinse, gel, varnish, SDF . Current water fluoridation target: 0.7 ppm .	Thinking fluoride is mainly pre-eruptive/systemic.	5
Understand the role of diet in caries	Diet contributes by providing fermentable carbohydrate substrate for biofilm acid production. Frequency, between-meal exposure, and sticky form increase risk; sugar substitutes, especially xylitol-containing products, may reduce cariogenic challenge when used alongside fluoride and hygiene.	Diet = total intake; nutrition = absorption/use of nutrients. Vipeholm: sticky sugar between meals is worst.	Saying all sugar exposure is equivalent.	6
Demonstrate understanding of caries risk assessment	CRA estimates disease probability from the balance of protective and risk/pathologic factors using history, exam, radiographs, CRA form, and saliva tests when indicated. The output drives recall, bitewing interval, fluoride/product choice, and prevention intensity.	Low 0; moderate 1-9; high 10+; extreme if high risk plus confirmed severe dry mouth/indicated culture logic. Recall/bitewings: low 12/24 , moderate 6/18 , high 4/12 , extreme 3/6 months.	Risk level from a single test; forgetting recall intervals.	8
Demonstrate familiarity with caries activity tests	Activity/risk tests are adjuncts used when history or symptoms suggest risk, especially salivary flow testing for dry mouth. They do not replace clinical judgment, examination, radiographs, and risk-factor assessment.	Stimulated saliva flow test; dry mouth on form if <0.7 ml/min ; bacterial culture may be used in extreme-risk logic.	Treating salivary/bacterial tests as stand-alone diagnostic truth.	7-8
Know efficacy of caries-preventive agents	Fluoride toothpastes, mouthrinses, gels, varnish, and SDF have source-supported caries-preventive roles. Varnish can remineralize early enamel caries; 38% SDF arrests dentine caries; xylitol gum has evidence of caries reduction but systematic reviews call for stronger trials.	5% NaF varnish; 38% SDF ; 5000 ppm prescription fluoride for high/root/xerostomia contexts.	Memorizing every brand instead of matching agent to indication and risk.	5, 9
Understand application of caries-preventive agents	Product use escalates with risk: low risk uses nutrition counseling and OTC fluoride toothpaste; moderate adds OTC 0.05% NaF rinse and varnish; high substitutes prescription 1.1% NaF/5000 ppm fluoride; extreme adds dry-mouth rinsing/buffers and salivary/dry-mouth management.	Varnish: small amount, aspiration, expectorate, avoid eating/drinking/smoking for 30 min. SDF : warn about black staining and contraindications.	Giving high-risk products to low-risk patients without reason.	9
Assess clinical cases and provide prevention strategies	Case management should identify risk level, surface, dentition, cavitation, saliva status, fluoride exposure, diet pattern, and ability to maintain hygiene. Choose prevention and non-restorative treatment before surgical restoration when appropriate.	Non-cavitated occlusal: sealant +/- 5% NaF varnish. Non-cavitated approximal/smooth: varnish or resin infiltration options. Cavitated: 38% SDF . Root: 5000 ppm fluoride or SDF .	Jumping straight to fillings; ignoring risk level and cavitation status.	9
Understand etiology and pathology of caries	Etiology is multifactorial: cariogenic biofilm, fermentable carbohydrate, susceptible host/tooth/saliva, and time. Pathology is repeated acid-mediated mineral dissolution followed by tissue breakdown if the surface integrity fails.	Acidogenic/chemoparasitic theory; Orland/Keys demonstrated bacteria are required; caries can be infectious/transmissible.	"Sugar alone causes caries."	1
Recognize stages of caries development	The clinical sequence is normal surface -> decalcification/hyper-white change -> staining/darker surface -> break in surface integrity -> cavitation and possible pulpal/periapical consequences.	White spot can arrest/reverse while surface remains intact; cavitation is later.	Calling every white spot a cavity.	2
Know Stephan curve	Stephan curve describes plaque pH drop after carbohydrate exposure and recovery over time; enamel begins dissolving around critical pH 5.5 .	Critical pH: 5.5 .	Memorizing the curve name without pH meaning.	1
Be familiar with recent concepts in	Modern cariology emphasizes DMD and CAMBRA : distribution,	CAMBRA = Caries Management By Risk	Treating management as	1, 8, 9

Syllabus objective	Exam-ready answer	Must-memorize details	Common wrong-answer trap	Chapter
caries management	management/treatment, determinants; risk-based prevention; patient participation; remineralization of non-cavitated lesions; minimal operative intervention for cavitated disease.	Assessment.	restoration-only.	
Recognize normal enamel/dentin/cementum vs caries	Sound tissues lack chalky demineralization, softness, cavitation, or root-surface discoloration. Enamel lesions begin as subsurface mineral loss with intact surface; dentin/root involvement becomes softer and more clinically destructive.	Clean/dry visual exam improves recognition.	Diagnosing from one glance on wet plaque-covered teeth.	2
Assess clinical stages and tooth-surface susceptibility	Lower first molars are highly susceptible because they erupt early, have pits/fissures, and sit farther from major salivary protection. Lower canines are least susceptible because they erupt later, are close to salivary glands, and have smooth surfaces.	High-risk sites: pits/fissures, proximal surfaces, cervical margins, exposed roots, restoration margins, surfaces near dentures/bridges.	Overvaluing fluoride for deep pits/fissures; fluoride protects smooth surfaces best.	2
Know pulpal clinical and histologic changes due to caries	Pulp responds to caries with immune, vascular, neurogenic, and morphologic changes. Bacterial by-products pass through dentinal tubules and cause inflammation before actual bacterial invasion. With progression, increased vascular permeability and tissue pressure create pain and may lead to necrosis/periapical spread.	Pulp is low-compliance, has inefficient lymph drainage in inflammation, and lacks collateral blood supply.	Thinking pulp stays normal until direct exposure.	3
Understand secondary decay and pulp responses	Recurrent/secondary caries around restorations can irritate pulp through the same microbial and chemical pathway as primary lesions; deeper lesions increase pulpal irritation risk because tubule number and diameter increase closer to pulp.	Recurrent caries = new disease around/beneath restoration.	Calling recurrent caries “the same old cavity coming back” without new disease activity.	2-3
Know systemic/topical fluoride modalities and fluorosis risk	Fluoride modalities include water, salt, milk, supplements, toothpaste, mouthrinse, gels, varnish, and SDF . Excess systemic intake during enamel development can cause dental fluorosis ; severe skeletal effects are linked to excessive exposure.	Fluorosis = hypomineralization/disrupted enamel development from excessive fluoride intake.	Forgetting fluoride benefit and fluorosis risk must be balanced in young children.	5
Know normal and abnormal salivary flow	Resting unstimulated flow is about 0.3-0.5 ml/min; stimulated mechanical flow about 1-2 ml/min; sour stimulation about 5-10 ml/min. Moderate xerostomia is around 0.2 ml/min; severe xerostomia is <0.1 ml/min. CRA dry mouth threshold uses <0.7 ml/min by saliva flow test.	Daily saliva 1-1.5 L; pH 6.0-7.5; 99% water, 1% solids.	Mixing resting flow values with CRA dry-mouth threshold.	7-8
Describe caries risk levels	Low risk has no risk points; moderate risk is 1-9 points; high risk is 10+ points; extreme risk is high-risk status with severe salivary/disease indicators as defined by the form logic.	Recall/bitewing: low 12/24 , moderate 6/18 , high 4/12 , extreme 3/6 .	Forgetting recall intervals even when products are known.	8-9

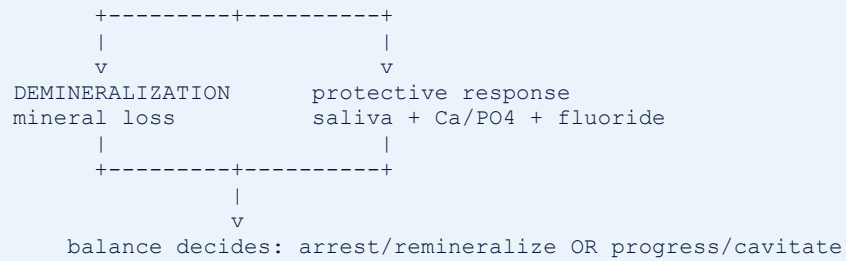
One-Page Disease Model [Back to TOC](#)

VISUAL PATHWAY: Core Framework

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BIOFILM + FERMENTABLE CARBOHYDRATE + TIME
|
v
organic acids in plaque
|
v
pH falls below critical range
|

```



ANCHOR

Caries is a balance problem, not just a bacteria problem. The exam move is to name the pathologic factors and protective factors, then decide how the balance changes the lesion and the management plan.

Module Briefs [Back to TOC](#)

Module 1. Caries as a Dynamic Disease Process [Back to TOC](#)

Exam Core

Dental caries is a chronic disease of enamel, dentin, and cementum caused by microorganisms acting on fermentable carbohydrate. The disease requires a susceptible host/tooth/saliva environment, cariogenic plaque/biofilm, carbohydrate substrate, and time. The clinical lesion reflects repeated pH-driven mineral loss when demineralization outruns remineralization.

Tested Vocabulary

Term	Exam definition	Why it matters	Do not confuse with
Dental caries	Localized destructive disease process of mineralized tooth tissues caused by microorganisms acting on fermentable carbohydrate.	Core course definition.	The cavity alone; caries is the process.
Biofilm/plaque	Thin bacterial colony on tooth surface.	Produces acids from carbohydrate substrate.	Food debris alone.
Demineralization	Mineral loss from enamel/tooth structure when pH fluctuates below critical level.	Explains white spot and lesion progression.	Cavitation; demineralization can occur before surface break.
Remineralization	Repair of partially demineralized enamel through recrystallization by mineral salts.	Explains lesion arrest/reversal potential.	Restoration/filling.
Stephan curve	Plaque pH response after carbohydrate exposure.	Links diet to acid challenge and critical pH.	A caries index.
Critical pH	Approximate pH at which enamel begins to dissolve.	Enamel critical pH emphasized by the professor.	Salivary pH range.
CAMBRA	Caries Management By Risk Assessment.	Modern management framework.	A diagnostic device.
DMD	Distribution, Management/Treatment, Determinants.	Explains modern changes in caries.	Doctor of dental medicine in this course context.

Mechanisms and Sequences

Trigger	Biologic/chemical event	Tooth/tissue result	Clinical implication
Fermentable carbohydrate exposure	Cariogenic bacteria produce organic acids	Plaque pH drops	Repeated acid challenges drive demineralization.
pH below critical threshold	Minerals leave enamel	Subsurface mineral loss	White spot/incipient lesion can occur before cavitation.
Remineralizing environment	Saliva, calcium, phosphate, fluoride support recrystallization	Lesion can arrest/remineralize if surface intact	Prevention can be definitive for non-cavitated lesions.
Continued imbalance	Demineralized tooth plus bacterial enzymes	Surface integrity breaks	Cavitation/decay; non-restorative vs restorative decision changes.

Clinical Recognition

Pattern	Meaning
White/chalky spot	Early enamel demineralization, potentially reversible/arrestable if surface intact.
Cavitation	Break in surface integrity; later-stage lesion.
Pit/fissure lesion	Common high-susceptibility morphology, especially molars.
Root lesion	Cementum/root surface involvement, often in exposed root context.

PROFESSOR SIGNAL

- Focus on: caries as a dynamic disease process, not a static hole.
- Why it matters: the professor repeatedly links bacteria, fermentable carbohydrate, acid, and mineral loss.
- Exam move: when asked "how caries develops," write the sequence: bacteria + fermentable carbohydrate + time -> acid -> pH drop -> demineralization -> possible cavitation.
- Focus on: critical pH.
- Why it matters: the professor highlighted the enamel dissolution threshold as a take-home point.
- Exam move: know 5.5 for enamel critical pH.

Common Confusions

Pair	Difference
Demineralization vs cavitation	Demineralization is mineral loss; cavitation is surface breakdown.
Caries process vs cavity	Caries is disease activity; cavity is one possible structural result.
Diet vs substrate	Diet is total intake; substrate is fermentable carbohydrate available to plaque.
CAMBRA vs CRA	CAMBRA is management by risk assessment; CRA is the risk assessment process/form.

EXAM TRAP

- A white spot is not automatically a cavity.
- Caries requires time; a single sugar exposure is not the whole disease model.
- Saliva is part of the host/protective system.
- Fluoride and saliva shift the balance toward remineralization.

MINI CHECKLIST

- [] Define caries using mineralized tissues + microorganisms + fermentable carbohydrate.
- [] Sequence acid production to demineralization to cavitation.
- [] State critical pH **5.5**.
- [] Explain demineralization vs remineralization.
- [] Name DMD and **CAMBRA**.
- [] Name key cariogenic bacteria: *S. mutans*, *S. sobrinus*, *L. acidophilus*, *A. viscosus*.

Module 2. Clinical Detection and Lesion Behavior [Back to TOC](#)

VISUAL PATHWAY: Clinical Diagnosis Decision Tree

```

clean + dry tooth
|
v
lesion present?
|
+-- no -> routine prevention + recall
|
+-- yes -> active vs arrested?
      |
      v
      cavitated?
      |         |
      no        yes
      |         |
      v         v
remineralize / seal / infiltrate  arrest with SDF or restore when indicated
    
```

Exam Core

Diagnosis is a decision process: identify whether disease is present, whether the lesion is active, whether it is cavitated, and whether management should be observe/prevent/fill. Official sources emphasize visual examination of clean, dry surfaces, careful tactile use, and radiographs; the professor warns against overreliance on one test.

Tested Vocabulary

Term	Exam definition	Why it matters	Do not confuse with
White spot lesion	Chalky white early sign of demineralization with intact surface.	Can be reversed/arrested.	Cavitated lesion.

Term	Exam definition	Why it matters	Do not confuse with
Arrested caries	Lesion that has stopped progressing; often shiny/lustrous, may be brown.	Indicates activity status.	Reversed white spot; arrested lesions are not always “gone.”
Recurrent caries	New caries around or beneath an existing restoration.	Common clinical classification.	The original lesion simply “returning.”
Root caries	Discrete, well-defined, discolored soft area on cementum/root surface; may extend to enamel.	Root exposure/high-risk patients.	Cervical enamel lesion.
Reliability	Same diagnosis repeatedly.	Test quality concept.	Validity.
Validity	Correct diagnosis.	Test quality concept.	Reliability.
Sensitivity	Ability to identify disease-positive cases.	Helps interpret screening/diagnostic tests.	Specificity.
Specificity	Ability to identify disease-free cases.	Helps interpret false positives.	Sensitivity.
PPV	Probability a positive result truly has disease.	Depends on test and disease context.	Sensitivity.
NPV	Probability a negative result truly lacks disease.	Depends on test and disease context.	Specificity.

Mechanisms and Sequences

1. Clean and dry the tooth surface.
2. Inspect visually under good lighting.
3. Use a systematic approach.
4. Use explorer/probe sparingly and sensibly: side of probe to remove biofilm, check surface texture and break, not forceful “stickiness” testing.
5. Add radiographs when indicated.
6. Decide lesion presence, activity, cavitation, and management.

Clinical Recognition

Clinical clue	Interpretation
Hyper-white/chalky area	Early demineralization/white spot.
Darker stained surface	May follow decalcification; stain alone does not prove active caries.
Break in surface integrity	Cavitation.
Shiny, lustrous lesion	Arrested lesion.
Brown stain	Can occur with arrested caries but is not required.
Soft discolored root area at CEJ/root	Root caries.

PROFESSOR SIGNAL

- Focus on: white spots can be reversed/arrested.
- Why it matters: the professor explicitly flags this as a take-home concept.
- Exam move: for intact white spot/non-cavitated lesions, think prevention/remineralization rather than automatic restoration.

- Focus on: visual exam first; tests are not perfect.
- Why it matters: the professor compared test concepts like sensitivity/specificity and warned against treating any test as 100% reliable.
- Exam move: answer diagnostic questions with combined judgment: visual exam + radiographs + risk context + careful tactile information.

Common Confusions

Pair	Difference
Active vs arrested	Active is progressing; arrested has stopped and often looks shiny/lustrous.
Initial vs gross/severe	Initial is early and often non-cavitated; gross/severe has substantial breakdown.
Probing vs sensible tactile exam	Forceful probing is destructive and inaccurate; sensible use checks texture/surface break.
Sensitivity vs PPV	Sensitivity asks how well positives are detected among diseased; PPV asks how often positive results are true positives.

EXAM TRAP

- Do not recommend aggressive explorer pressure into demineralized enamel.
- Do not diagnose caries from stain alone.
- Do not ignore tooth susceptibility: lower first molar is high risk; lower canine is low risk.
- Do not skip drying/cleaning before visual diagnosis.

MINI CHECKLIST

- [] Explain why lower first molars are highly susceptible.
- [] Explain why lower canines are relatively protected.
- [] Define reliability, validity, sensitivity, specificity, PPV, NPV.
- [] Identify white spot, arrested, recurrent, and root caries.
- [] State why probing can damage demineralized areas.

Module 3. Dentinal Involvement, Pulpal Response, and Pain [Back to TOC](#)

VISUAL PATHWAY: Pulpal Progression Map

```
deepening caries
|
v
bacterial by-products diffuse through dentinal tubules
|
v
pulp inflammation in low-compliance space
|
+-- short sharp cold pain -> reversible pulpitis pattern
|
+-- lingering/spontaneous pain -> irreversible pulpitis pattern
```

|
+-- loss of vitality/periapical spread -> necrosis pathway

Exam Core

Caries affects pulp before bacteria must physically invade it. Bacterial by-products diffuse through dentinal tubules, and as the lesion approaches pulp, increased tubule number/diameter increases irritation risk. The pulp is trapped in rigid dentin, so inflammation raises tissue pressure, compresses vessels, worsens circulation, causes pain, and may progress to necrosis/periapical disease.

Tested Vocabulary

Term	Exam definition	Why it matters	Do not confuse with
A-delta fibers	Myelinated fibers; sharp/pricking/cold pain; faster conduction; low threshold.	Cold testing and sharp pain.	C fibers.
C fibers	Unmyelinated fibers; burning/aching/lingering pain; slower conduction; high threshold.	Lingering inflammatory pain.	A-delta fibers.
Reversible pulpitis	Mild inflammation that can resolve if cause removed.	Treatment is remove cause/insulate.	Irreversible pulpitis.
Irreversible pulpitis	Moderate/severe inflammation that will not resolve after stimulus removal.	Treatment is RCT or extraction.	Reversible pulpitis.
Necrosis	Pulp death; often asymptomatic; rate increases in closed/sealed teeth.	Requires endodontic management.	Irreversible pulpitis.
Hyperplastic pulpitis	Irreversible pulpitis with chronically inflamed young pulp growing into decayed occlusal surface.	Reddish cauliflower-like growth.	Gingival tissue.
Internal resorption	Dentinoclastic destruction/enlargement of root canal space.	Needs immediate pulpectomy/RCT.	External resorption.
Calcific metamorphosis	Canal obliteration response often after trauma.	May cause false negative vitality tests.	Normal secondary dentin only.

Mechanisms and Sequences

1. Caries progresses through enamel/dentin.
2. Carious dentin/enamel contain many bacteria, but deeper layers have fewer bacteria.
3. Toxic bacterial by-products pass through dentinal tubules.
4. Local chronic inflammatory infiltrate develops.
5. As decay approaches pulp, tubule number and diameter increase, raising irritation potential.
6. Inflammatory mediators cause pain directly by lowering nerve threshold and indirectly through vasodilation/permeability/tissue pressure.
7. Low-compliance pulp space + inefficient lymph drainage + no collateral blood supply can lead to vessel compression, circulatory collapse, necrosis, and periapical spread.

Clinical Recognition

Symptom/test pattern	Likely concept
Brief sharp cold sensitivity	A-delta fiber response; may fit reversible inflammation depending context.
Lingering pain after stimulus	Irreversible pulpitis pattern.
Aching/burning pain	C fiber contribution.

Symptom/test pattern	Likely concept
Asymptomatic nonvital pulp	Necrosis can be asymptomatic.
Cold relieves pain in severe cases	Can occur in irreversible pulpitis.
Reddish tissue mass into carious occlusal lesion	Hyperplastic pulpitis.

PROFESSOR SIGNAL

- Focus on: deeper caries means more pulpal irritation risk.
- Why it matters: the professor emphasizes that tubules become more numerous/larger closer to pulp, giving irritants more access.
- Exam move: connect lesion depth to dentinal tubules and pulpal inflammatory response.
- Focus on: A-delta vs C fiber pain quality.
- Why it matters: the professor highlighted cold/sharp vs gnawing/aching pain.
- Exam move: sharp cold pain = A-delta; burning/aching/lingering = C fiber/inflammatory pain.

Common Confusions

Pair	Difference
Reversible vs irreversible pulpitis	Reversible resolves after cause removed; irreversible does not and needs RCT/extraction.
Bacterial invasion vs by-product diffusion	Pulp can be affected by by-products before bacterial invasion.
A-delta vs C fibers	A-delta fast/sharp/cold; C slow/burning/aching.
Necrosis vs calcification	Necrosis is death; calcification can obstruct canal and vitality testing.

EXAM TRAP

- Do not say the pulp is unaffected until direct exposure.
- Do not ignore the rigid dentin-wall pressure problem.
- Do not recommend a filling-only answer for irreversible pulpitis.
- Do not forget water coolant prevents heat injury during dentin cutting.

MINI CHECKLIST

- [] List pulp functions: inductive, formative, nutritive, defensive, sensory.
- [] Explain A-delta vs C fibers.
- [] Sequence lesion progression to necrosis.
- [] Distinguish reversible pulpitis, irreversible pulpitis, necrosis.
- [] Explain why pulpal inflammation is dangerous in a rigid chamber.

Module 4. Epidemiology, Distribution, and Determinants [Back to TOC](#)

Exam Core

Distribution is about who gets caries, where lesions occur, and how severity is measured. The exam distinction is **prevalence/incidence vs severity/intensity indices** and **total caries vs untreated caries**. Determinants include age, SES, race/ethnicity, fluoride exposure, diet, saliva, plaque, and protective behaviors.

Tested Vocabulary

Term	Exam definition	Why it matters	Do not confuse with
Prevalence	Existing number of cases.	Population burden.	DMFT.
Prevalence rate	Percent with disease.	Rate/proportion.	Incidence rate.
Incidence	New cases.	Disease onset over time.	Prevalence.
Total caries experience	Treated + untreated caries.	Broad history of disease.	Untreated caries only.
Untreated caries	Disease not yet treated.	Professor emphasized this distinction.	Filled/restored disease.
DMFT/DMFS	Decayed, missing due to caries, filled teeth/surfaces in permanent dentition.	Severity/intensity index.	Prevalence.
dmft/dmfs	Primary-tooth analog using lowercase.	Primary dentition index.	Permanent indices.
deft/defs	Primary teeth/surfaces with decay or extraction indicated; missing generally avoided because exfoliation confounds cause.	Primary dentition nuance.	dmft if missing cause unknown.

Mechanisms and Sequences

Distribution level	What to know
National	NHANES.
State/local	State health departments, ASTDD, local surveys.
International	Review papers and WHO data.
Subgroups	Age, gender, race/ethnicity, income/SES.
Tooth/surface	Smooth-surface proportion declined; pit-and-fissure proportion increased.

Clinical Recognition

Pattern	Exam meaning
Caries increases with age	Both treated and untreated burden rise across adult age groups.
Lower SES	Higher untreated caries and worse access/risk profile.
Youth total caries vs untreated caries	Hispanic youth highest total caries; African-American youth highest untreated caries in emphasized source framing.
Smooth vs pit/fissure shift	Smooth-surface proportion declined; pit-and-fissure proportion increased.

**PROFESSOR
SIGNAL**

- Focus on: **DMFT/DMFS** is severity/intensity, not prevalence.
- Why it matters: the professor repeats this distinction.
- Exam move: if the question asks prevalence, answer with percent/people; if it asks DMFT, answer severity/intensity.
- Focus on: untreated caries matters.
- Why it matters: the professor specifically downplayed raw prevalence-number memorization and emphasized untreated patterns.
- Exam move: know high-level trends and subgroup distinctions, not every percentage.

Common Confusions

Pair	Difference
Total caries vs untreated caries	Total includes treated/filled; untreated is current untreated decay.
Prevalence vs DMFT	Prevalence is presence of disease in people; DMFT is severity/intensity count.
Permanent vs primary notation	Permanent uses uppercase; primary uses lowercase.
Missing in primary dentition	Missing due to exfoliation may not mean caries, so deft/defs can be used.

EXAM TRAP

- Do not memorize every distribution percentage unless the professor explicitly says to.
- Do not call total caries the same thing as untreated caries.
- Do not treat gender as a major current difference; slides say male-female differences are no longer significant.
- Do not ignore SES.

MINI CHECKLIST

- [] Define prevalence, prevalence rate, incidence, incidence rate.
- [] Explain total caries vs untreated caries.
- [] Explain **DMFT/DMFS/dmft/dmfs/deft/defs**.
- [] State the tooth-surface shift.
- [] Name high-risk population patterns without overfocusing on raw numbers.

Module 5. Fluoride Systems and Remineralization Strategy [Back to TOC](#)

VISUAL PATHWAY: Fluoride Action Loop

```

acid challenge lowers plaque pH
      |
      v
fluoride available in plaque/saliva/tooth surface
    
```

```

|
+-- reduces demineralization
|
+-- enhances remineralization
|
v
lesion is more likely to arrest before cavitation

```

Exam Core

Fluoride is central to caries prevention. Official sources cover systemic and topical modalities, historical fluoridation, **fluorosis**, and product choices. The professor emphasizes water fluoridation definition, topical mechanisms, fluoride products by risk level, **SDF** side effects, and recall/product recommendations.

Tested Vocabulary

Term	Exam definition	Why it matters	Do not confuse with
Systemic fluoride	Ingested fluoride delivered through water, salt, milk, supplements.	Population-level exposure; fluorosis risk.	Topical fluoride.
Topical fluoride	Fluoride applied/available at tooth surface: toothpaste, rinse, gel, varnish, SDF .	Main clinical preventive mechanism.	Systemic fluoride only.
Water fluoridation	Controlled addition of fluoride compound to public water supply to best prevent dental caries while avoiding unsightly fluorosis .	Professor emphasis says definition is exam-relevant.	Natural fluoride contamination.
Dental fluorosis	Hypomineralization and disruption of enamel development due to excessive fluoride intake.	Side effect/risk balance.	Cariou white spot.
Osteosclerosis	Abnormal hardening of bone with increased skeletal density and little disturbance of remodeling.	Fluoride toxicity concept.	Dental fluorosis .
APF	Acidulated phosphate fluoride; acidic environment increases enamel fluoride uptake.	Professional topical option.	Neutral NaF.
SDF	Silver diamine fluoride; high fluoride, antimicrobial silver, arrests carious lesions.	Major non-restorative agent.	Fluoride varnish.

Mechanisms and Sequences

Fluoride action	Meaning
Reduces demineralization	Makes tooth mineral less soluble during acid challenge.
Promotes remineralization	Helps mineral redeposit in early lesions.
Acts predominantly on smooth surfaces	Professor emphasis and slides emphasize smooth-surface protection.
Plaque fluoride release	Plaque contains bound fluoride; pH drop can shift bound fluoride to ionic form.
Preventive risk balance	Children under 6 require fluoride benefit vs fluorosis risk consideration.

Clinical Recognition

Modality	Key details
Water	Current recommendation 0.7 ppm ; early trials included Grand Rapids/Muskegon, Newburgh/Kingston, Evanston/Oak Park, Brantford/Sarnia.
Toothpaste	Daily 900-1500 ppm fluoride; OTC examples include 1000-1500 ppm.
Mouthrinse	0.05% NaF daily; 0.2% NaF weekly; 0.1% APF weekly.
In-home gel	1.1% NaF or 0.4% SnF2 daily by brush/custom tray.
Professionally applied gel	2% NaF, 1.23% APF, 8% SnF2.
Varnish	5% NaF varnish, 22,600 ppm; slows progression; high contact time.
SDF	~44,500 ppm F; silver antimicrobial, ammonia solvent, fluoride remineralization.

PROFESSOR SIGNAL

- Focus on: definition of water fluoridation.
- Why it matters: professor explicitly says it will be on the exam.
- Exam move: define controlled addition to public water supply to prevent caries while avoiding **fluorosis**.
- Focus on: product choice by risk level.
- Why it matters: professor says students need to know what products to use for high vs low risk.
- Exam move: low -> OTC fluoride toothpaste; moderate -> add OTC 0.05% rinse/varnish; high -> prescription **1.1% NaF/5000 ppm**; extreme -> dry-mouth/buffer strategies.
- Focus on: **SDF** disadvantages.
- Why it matters: professor says patients must be warned before application.
- Exam move: know black staining, silver allergy, mucosal ulceration/stomatitis cautions, mild gastric irritation, taste/tissue issues.

Common Confusions

Pair	Difference
NaF vs APF	NaF is neutral/stable/better tolerated; APF is acidic and increases enamel fluoride uptake but may affect materials differently.
SnF2 vs NaF	SnF2 can help sensitivity but has metallic taste, staining, tissue irritation, instability.
Varnish vs SDF	Varnish remineralizes early enamel caries; SDF arrests dentine caries and stains lesions black.
Fluorosis vs white spot	Fluorosis is developmental hypomineralization from excess fluoride; white spot caries is demineralization from plaque acid.

EXAM TRAP

- Do not memorize chemical formulas that professor says are not tested.
- Do not ignore **0.7 ppm** current water fluoridation level.
- Do not forget smooth surfaces are where fluoride acts best.
- Do not forget **SDF** is not cosmetic; black staining is expected.

MINI CHECKLIST

- [] Define water fluoridation.
- [] Separate systemic vs topical fluoride.
- [] State current water fluoridation level.
- [] Know fluoride toothpaste/rinse/gel/varnish/**SDF** roles.
- [] Know **fluorosis** definition.
- [] Know **SDF** benefits, indications, and side effects.

Module 6. Diet, Sugar Exposure, and Caries Ecology [Back to TOC](#)

Exam Core

Diet contributes substrate for acid production. The strongest exam message is not simply “sugar is bad,” but that caries risk rises with fermentability, sticky form, frequency, and between-meal exposure. Sugar substitutes, especially xylitol, can reduce risk but do not replace fluoride, oral hygiene, and risk-based care.

Tested Vocabulary

Term	Exam definition	Why it matters	Do not confuse with
Diet	Total intake of substances that produce nourishment and energy.	Syllabus objective.	Nutrition.
Nutrition	Absorption/use of nutrients.	Distinction tested.	Diet.
Disaccharides	Sucrose, lactose, maltose.	Fermentable carbohydrate categories.	Monosaccharides.
Monosaccharides	Glucose, fructose.	Sugar classification.	Disaccharides.
Sucrose	Glucose + fructose.	Important cariogenic sugar.	Xylitol.
Xylitol	Sugar alcohol/sugar substitute with antibacterial and low plaque-pH-drop properties.	Preventive product.	Sucrose.

Mechanisms and Sequences

Evidence pattern	Exam meaning
Wartime sugar restriction	Lower sugar availability associated with lower caries.
Vipeholm	Sticky sugar between meals greatly increases caries activity.
Turku xylitol	Xylitol group had lowest/no caries in cited trial framing.
Hereditary fructose intolerance	Lower caries prevalence observed because sucrose/fructose intake limited.
Soft drinks	Positive association with caries vs water.

Clinical Recognition

Diet situation	Risk interpretation
----------------	---------------------

Diet situation	Risk interpretation
Sticky sugar between meals	Highest cariogenic pattern.
Sugar at meals only	Lower than prolonged between-meal exposure.
Frequent sipping sugary/carbonated drinks	High risk due to repeated acid challenges.
Xylitol gum/products with fluoride routine	Potentially protective adjunct.
Absence of refined sugar	Does not guarantee no caries.

PROFESSOR SIGNAL

- Focus on: Vipeholm logic and sticky between-meal sugar.
- Why it matters: professor signals diet examples and numbers were used to make the point memorable, not just decorative.
- Exam move: if asked what diet pattern is worst, answer sticky fermentable carbohydrate between meals.
- Focus on: diet is a determinant, not the only cause.
- Why it matters: professor ties diet back to bacteria fermenting carbohydrates and producing acids.
- Exam move: always connect diet to plaque acid and pH, not just calories.

Common Confusions

Pair	Difference
Diet vs nutrition	Diet = intake; nutrition = absorption/use.
Sucrose vs xylitol	Sucrose is fermentable glucose + fructose; xylitol is a sugar alcohol with anticaries effects.
Frequency vs amount	Frequent between-meal exposures are especially damaging because pH repeatedly drops.
Sugar-free gum evidence	Helpful but systematic reviews still call for stronger trials; use as adjunct.

EXAM TRAP

- Do not say malnutrition has strong evidence as a caries determinant; sources say weak at best.
- Do not say genetic/familial aggregation is proven; sources say inconclusive.
- Do not ignore individual variation in caries under uniform diet conditions.
- Do not forget xylitol side effects: laxative effect, palpitations, fatal to dogs.

MINI CHECKLIST

- [] Define diet vs nutrition.
- [] List mono/disaccharides and sucrose composition.
- [] Explain Vipeholm conclusions.
- [] Explain xylitol action.
- [] Explain why frequency/stickiness matters.

Module 7. Saliva, Xerostomia, and Activity Testing [Back to TOC](#)

Exam Core

Saliva protects against caries through clearance, buffering, calcium/phosphate delivery, remineralization support, antimicrobial proteins, and mucosal protection. Dry mouth shifts the caries balance strongly toward disease, especially root/cervical/radiation patterns. Flow testing is an adjunct when symptoms, history, medications, or risk form logic indicate it.

Tested Vocabulary

Term	Exam definition	Why it matters	Do not confuse with
Xerostomia	Dry mouth; clinically important low-saliva state.	High-risk factor.	Normal transient thirst.
Hyposalivation	Reduced salivary production/flow.	Objective testing concept.	Subjective dry-mouth complaint.
Parotid	Major gland producing watery saliva.	Dominates stimulated mechanical flow.	Submandibular.
Submandibular	Major gland producing viscous saliva; major resting contributor.	Resting saliva contribution.	Parotid.
Sublingual/minor glands	Viscous saliva.	Lubrication/protection.	Parotid watery saliva.
IgA	Salivary immunoglobulin that inhibits adhesion.	Professor-emphasized immune component.	IgG/IgM phagocytosis roles.
Sjogren syndrome	Autoimmune exocrine-gland disorder with dry mouth/dry eyes/difficulty swallowing.	Professor emphasis flagged as important.	Simple medication dry mouth.

Mechanisms and Sequences

Protective saliva function	Mechanism
Clearance	Clears food and bacterial debris; reduces carbohydrate exposure time.
Buffering	Bicarbonate/phosphate regulate pH.
Mineral support	Calcium and phosphate inhibit demineralization and support remineralization.
Pellicle/protein support	Mucins, proline-rich proteins, statherin, glycoproteins.
Antimicrobial	Lysozyme, lactoperoxidase, lactoferrin, histatin, cystatin, agglutinin, mucin, immunoglobulins.
Immune	IgA inhibits adhesion; IgG and IgM increase phagocytosis.

Clinical Recognition

Value/pattern	Meaning
1-1.5 L/day	Daily saliva production.
99% water, 1% solids	Basic composition.
pH 6.0-7.5	Normal salivary pH range.
0.01-0.03 ppm fluoride	Salivary fluoride range from slides.

Value/pattern	Meaning
Resting 0.3-0.5 ml/min	Normal unstimulated flow.
Mechanical stimulated 1-2 ml/min	Normal mechanical stimulation.
Sour stimulated 5-10 ml/min	Strong taste stimulation.
Moderate xerostomia 0.2 ml/min	Low flow.
Severe xerostomia <0.1 ml/min	Very low flow.
CRA dry mouth <0.7 ml/min	Form threshold by salivary flow test.

**PROFESSOR
SIGNAL**

- Focus on: salivary functions.
- Why it matters: professor says there will be questions about salivary functions.
- Exam move: list clearance, buffering, calcium/phosphate/reminerization, antimicrobial action, mucosal protection.
- Focus on: **Sjogren** syndrome and medications causing low saliva.
- Why it matters: professor calls **Sjogren** a major topic and says students should know medication categories.
- Exam move: dry mouth + dry eyes + swallowing difficulty + autoimmune/exocrine = **Sjogren**; meds include antidepressants, antipsychotics, tranquilizers, antihistamines, anticholinergics, antihypertensives.
- Focus on: IgA.
- Why it matters: professor says the main antibody point is IgA.
- Exam move: IgA inhibits adhesion.

Common Confusions

Pair	Difference
Resting vs stimulated flow	Resting 0.3-0.5 ml/min; stimulated mechanical 1-2 ml/min; sour 5-10 ml/min.
Subjective xerostomia vs measured flow	Complaint triggers testing; risk form uses measured threshold.
IgA vs IgG/IgM	IgA inhibits adhesion; IgG/IgM increase phagocytosis.
Bicarbonate/phosphate vs calcium/phosphate	Bicarbonate/phosphate buffer pH; calcium/phosphate support mineral balance.

EXAM TRAP

- Do not ignore medication history in dry-mouth patients.
- Do not mix the salivary-flow values.
- Do not forget radiation/chemo as high-risk dry-mouth context.
- Do not overfocus on antiviral/antifungal lists if the exam emphasis is antimicrobial/caries protection.

MINI CHECKLIST

- [] State normal salivary flow rates.
- [] State dry-mouth thresholds.
- [] List saliva protective functions.
- [] Name salivary antimicrobial proteins.
- [] Explain **Sjogren** syndrome.
- [] Know IgA inhibits adhesion.

Module 8. Caries Risk Assessment and Recall Planning [Back to TOC](#)

VISUAL PATHWAY: Risk-to-Recall Ladder

LOW risk -> recall 12 mo -> bitewings 24 mo
MODERATE risk -> recall 6 mo -> bitewings 18 mo
HIGH risk -> recall 4 mo -> bitewings 12 mo
EXTREME risk -> recall 3 mo -> bitewings 6 mo

exam logic: risk level -> interval -> fluoride/product intensity

Exam Core

Caries risk assessment is the interaction/balance between protective and risk factors. It is not a single test. The form uses contributing conditions, general health, and clinical findings to assign low/moderate/high/extreme risk, then drives recall interval, bitewing interval, and product recommendations.

Tested Vocabulary

Term	Exam definition	Why it matters	Do not confuse with
Risk	Probability that a disease/event will occur.	Basic CRA definition.	Current disease.
Risk factor	Attribute/exposure associated with increased probability of disease.	CRA scoring.	Disease indicator.
Protective factor	Attribute/exposure associated with decreased probability of disease.	Balances risk.	Treatment after disease only.
CRA	Caries risk assessment.	Determines prevention and recall.	Caries diagnosis alone.
Extreme risk	High-risk patient with additional severe salivary/disease indicators per form logic.	More intensive recall/products.	Simply “very high points” only.

Mechanisms and Sequences

1. Take medical, dental, and social history.
2. Perform clinical exam.
3. Review radiographs.
4. Complete CRA form.
5. If dry-mouth complaint or meds reducing saliva, measure salivary flow.
6. If indicated by form logic, use bacterial culture.

7. Assign risk level.

8. Match recall interval, bitewing interval, and prevention plan.

Clinical Recognition

Risk level	Point/logic	Recall	Bitewings	Core advice
Low	0 points	12 months	24 months	Nutrition counseling; OTC fluoride toothpaste 2x/day.
Moderate	1-9 points	6 months	18 months	Low regimen + OTC 0.05% NaF rinse daily + 5% NaF varnish.
High	10+ points	4 months	12 months until no lesions	Moderate regimen + prescription 1.1% NaF/5000 ppm fluoride toothpaste 2x/day.
Extreme	High-risk plus severe salivary/disease logic	3 months	6 months until no lesions	High regimen + dry- mouth rinsing, baking soda gum, acid-buffering rinses.

PROFESSOR SIGNAL

- Focus on: do not decide risk from one test.
- Why it matters: professor emphasis repeats that decisions are based on a wealth of information.
- Exam move: answer CRA questions as combined history + exam + radiographs + form + saliva tests if indicated.
- Focus on: **36-month** time frame.
- Why it matters: professor tells students to remember the 3-year window.
- Exam move: use last 36 months for new lesions/restorations and missing teeth due to caries.
- Focus on: recall intervals.
- Why it matters: professor explicitly says recall intervals are needed for the exam.
- Exam move: memorize low **12/24**, moderate **6/18**, high **4/12**, extreme **3/6**.

Common Confusions

Pair	Difference
Low vs moderate	Low has protective profile and no points; moderate has 1-point factors.
Moderate vs high	High has 10-point factors like visible lesions, 3+ recent lesions/restorations, dry mouth, chemo/radiation.
Dry-mouth complaint vs high-risk dry mouth	Complaint triggers flow testing; <0.7 ml/min makes Q yes/high risk.
Recall vs bitewings	Recall is visit frequency; bitewings are radiographic interval.

EXAM TRAP

- Do not miss fluoride exposure as protective.
- Do not forget dental home as protective.
- Do not miss frequent/prolonged between-meal sugar exposure as high risk.
- Do not forget exposed roots/appliances/visible plaque are moderate risk factors.
- Do not skip salivary flow test when dry-mouth history is present.

MINI CHECKLIST

- [] Define risk, risk factor, protective factor.
- [] Explain CRA as risk/protective balance.
- [] Memorize risk point levels and recall/bitewing intervals.
- [] Know form categories A-Q.
- [] Know dry-mouth threshold **<0.7 ml/min**.
- [] Know the **36-month** window.

Module 9. Preventive Products and Case-Based Management [Back to TOC](#)

VISUAL PATHWAY: Product Escalation Ladder

low risk
 -> OTC fluoride toothpaste + diet/oral hygiene counseling
 moderate risk
 -> add OTC 0.05% NaF rinse + varnish
 high risk
 -> prescription 1.1% NaF / 5000 ppm fluoride + shorter recall
 extreme risk
 -> high-risk plan + dry-mouth/saliva management

Exam Core

The course tests whether students can match lesion type, risk level, and product. Management includes prevention, non-restorative treatment, and later restoration/filling. The strongest exam actions are fluoride risk-level recommendations, **SDF**/varnish/resin infiltration decisions, and recall-product plans.

Tested Vocabulary

Term	Exam definition	Why it matters	Do not confuse with
Non-restorative treatment	Caries management without conventional filling; includes sealants, fluoride, SDF , resin infiltration.	Explicit case-management objective.	No treatment.
Sealant	Non-restorative option especially for non-cavitated occlusal lesions.	Occlusal case pattern.	Resin infiltration.
Resin infiltration	Micro-invasive resin barrier for incipient/smooth/approximal enamel lesions.	Non-cavitated approximal/smooth surface option.	Composite restoration.
5% NaF varnish	Professional fluoride varnish for remineralization/early lesions.	Moderate/high risk and non-cavitated lesion	SDF .

Term	Exam definition	Why it matters	Do not confuse with
		management.	
38% SDF	Silver diamine fluoride used to arrest cavitated/dentine/root lesions.	Cavitated case pattern.	Varnish.
5000 ppm fluoride	Prescription high-fluoride toothpaste/gel.	High-risk, root caries, xerostomia/rampant caries context.	OTC toothpaste.
MI Paste/Recaldent	Casein-derived calcium/phosphate remineralization product; MI Paste Plus includes fluoride.	Xerostomia/high-risk adjunct.	Fluoride varnish.

Mechanisms and Sequences

Clinical case	Non-cavitated management	Cavitated management
Occlusal primary tooth	Sealant + 5% NaF varnish OR sealant alone	38% SDF
Approximal primary tooth	5% NaF varnish OR resin infiltration OR resin infiltration + 5% NaF varnish	38% SDF
Facial/lingual primary tooth	5% NaF varnish OR 1.23% APF	38% SDF
Occlusal permanent tooth	Sealant + 5% NaF varnish OR sealant alone	38% SDF
Approximal permanent tooth	5% NaF varnish OR resin infiltration OR resin infiltration + 5% NaF varnish	38% SDF
Facial/lingual permanent tooth	5% NaF varnish OR 1.23% APF	38% SDF
Root surface permanent tooth	5000 ppm/1.1% NaF toothpaste or gel; SDF	5000 ppm/1.1% NaF toothpaste or gel; SDF

Clinical Recognition

Product	Use	Watch-outs
OTC fluoride toothpaste	Low risk baseline; twice daily.	Does not replace high-risk Rx fluoride.
OTC 0.05% NaF rinse	Moderate-risk adjunct; after brushing.	Not for every low-risk patient by default.
Rx 0.2% NaF rinse	High-risk option.	Follow prescription regimen.
1.1% NaF / 5000 ppm	High risk, root caries susceptibility, rampant caries, xerostomia.	Professor emphasis says know 5000 ppm prescription toothpaste.
Chlorhexidine	Antimicrobial; effective against mutans streptococci, limited against lactobacilli.	Staining, dry mouth, unpleasant/change in taste, irritation, rare allergy.
Iodine	Broad antimicrobial; direct contact, allergy consideration.	Adjunct, not core fluoride replacement.
Xylitol	Sugar substitute; salivary stimulation and plaque effects.	Laxative effects/palpitations; fatal to dogs.
Baking soda/buffers	Acid buffering, dry-mouth support.	Adjunct; evidence/support varies.
Artificial saliva/xerostomia rinses	Dry-mouth symptom management.	Does not replace fluoride/risk management.
Recaldent/MI Paste	Calcium/phosphate remineralization support; milk/casein origin.	Contraindication/caution for milk protein/lactose-related issues

Product	Use	Watch-outs
		from guide framing.

PROFESSOR SIGNAL

- Focus on: products by risk level.
- Why it matters: professor says students need to know low/moderate/high recommendations and recall intervals.
- Exam move: risk level -> recall interval -> bitewing interval -> fluoride/product intensity.
- Focus on: resin infiltration.
- Why it matters: professor flags it as used for incipient/white lesions and smooth surfaces.
- Exam move: resin infiltration = micro-invasive, incipient, smooth/approximal, barrier/arrest, not cavitated bulk restoration.
- Focus on: fillings are not the target.
- Why it matters: professor says fillings/restorations are not the exam focus because students have not started that work.
- Exam move: answer with prevention/non-restorative strategies unless the question clearly asks later restorative care.

Common Confusions

Pair	Difference
Varnish vs SDF	Varnish remineralizes/slows early lesions; SDF arrests dentine/cavitated lesions and stains.
Sealant vs resin infiltration	Sealant for occlusal pits/fissures; resin infiltration for incipient smooth/approximal lesions.
Low-risk advice vs high-risk advice	Low = OTC toothpaste; high = prescription fluoride and shorter recall.
Antimicrobial vs fluoride	Antimicrobials reduce bacteria; fluoride shifts mineral balance.

EXAM TRAP

- Do not give a restoration answer when the source case asks for non-restorative management.
- Do not forget cavitated cases repeatedly point to **38% SDF**.
- Do not forget root surface cases point to **5000 ppm** fluoride/**SDF**.
- Do not omit side effects when recommending **SDF** or chlorhexidine.
- Do not memorize all product brand names at the expense of indication.

MINI CHECKLIST

- [] Match occlusal non-cavitated lesions to sealants +/- varnish.
- [] Match approximal/smooth non-cavitated lesions to varnish/resin infiltration.
- [] Match cavitated lesions to **SDF**.
- [] Match root lesions to **5000 ppm** fluoride/**SDF**.
- [] Escalate prevention by risk level.
- [] State **SDF** and chlorhexidine side effects.

Comparison Tables for Fast Discrimination [Back to TOC](#)

Sound / Initial / Cavitated / Arrested Lesions [Back to TOC](#)

Feature	Sound surface	Initial/white spot	Cavitated lesion	Arrested lesion
Surface integrity	Intact	Intact	Broken	Usually intact/stable surface
Appearance	Normal enamel/dentin/cementum	Chalky/hyper-white	Surface breakdown, cavity/decay	Shiny/lustrous; brown stain may occur
Disease activity	None	Active or potentially active early demineralization	Advanced active/destructive disease unless arrested	Stopped progression
Management logic	Maintain prevention	Remineralize/prevent; monitor	Non-restorative or restorative depending case	Monitor/prevention; not “reverse to normal”

Primary / Recurrent / Root / Rampant Caries [Back to TOC](#)

Type	Defining feature	High-yield clue	Management implication
Primary caries	New lesion on previously unrestored surface	Pits/fissures, proximal, smooth/cervical	Risk-based prevention/treatment.
Recurrent caries	New disease around/beneath restoration	Restoration margins/open contacts/food impaction	Evaluate restoration plus caries risk.
Root caries	Soft discolored root/cementum lesion, often near CEJ	Exposed root, xerostomia, older/high-risk patient	5000 ppm fluoride and/or SDF emphasized.
Rampant caries	Rapid widespread disease	Multiple lesions, high-risk ecology	Intensive risk reduction and high fluoride.

Risk Factor / Protective Factor / Disease Indicator [Back to TOC](#)

Category	Meaning	Examples
Risk factor	Increases probability of caries	Frequent sugar drinks/snacks, low fluoride, visible plaque, exposed roots, appliances, dry mouth, medication-induced low saliva.
Protective factor	Decreases probability of caries	Saliva flow/components, fluoride, good oral hygiene, restricted carbohydrate exposure, sugar substitutes, dental home.
Disease indicator	Evidence disease has occurred/is occurring	Cavitated/non-cavitated lesions, restorations due to caries, teeth missing due to caries, white spot lesions.

Systemic vs Topical Fluoride [Back to TOC](#)

Feature	Systemic fluoride	Topical fluoride
Route	Ingested and distributed	Applied/available at tooth surface
Examples	Water, salt, milk, supplements	Toothpaste, mouthrinse, gels, varnish, SDF
Main exam value	Population prevention; fluorosis balance	Direct caries prevention through plaque/saliva/tooth surface
Risk	Excess intake during development -> fluorosis	Product-specific side effects, swallowing risk in young children
Professor emphasis	Water fluoridation definition/current level	Mechanisms, product choice, SDF effects

Self-Applied vs Professional Fluoride [Back to TOC](#)

Feature	Self-applied	Professionally applied
Products	Toothpaste, mouthrinse, gels	Gel/foam, varnish, SDF
Typical users	Patient/caregiver at home	DDS/DMD, RDH, DT
Examples	OTC 0.05% NaF rinse; Rx 0.2% rinse; 1.1% NaF/5000 ppm paste	2% NaF, 1.23% APF, 5% NaF varnish, 38% SDF
Exam move	Match to risk level and adherence	Match to lesion/risk and side effects

Salivary Flow and Function [Back to TOC](#)

Item	Value / role
Daily saliva	1-1.5 L
Composition	99% water; 1% solids
pH	6.0-7.5
Fluoride	0.01-0.03 ppm
Resting unstimulated	0.3-0.5 ml/min
Mechanical stimulated	1-2 ml/min
Sour stimulated	5-10 ml/min
Moderate xerostomia	0.2 ml/min
Severe xerostomia	<0.1 ml/min
CRA dry-mouth threshold	<0.7 ml/min by salivary flow test

Pulp Pain Fibers
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Feature	A-delta fibers	C fibers
Myelin	Yes	No
Pain quality	Sharp, pricking, cold	Burning, aching, lingering
Diameter	1-5 micrometers	0.4-1 micrometers
Speed	6-30 m/sec	0.5-2 m/sec
Threshold	Low	High
Exam clue	Quick cold/sharp response	Persistent gnawing/aching inflammatory pain

Non-Restorative Treatment Matrix
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Surface/dentition	Non-cavitated	Cavitated
Occlusal primary/permanent	Sealant + 5% NaF varnish OR sealant alone	38% SDF
Approximal primary/permanent	5% NaF varnish OR resin infiltration OR both	38% SDF
Facial/lingual primary/permanent	5% NaF varnish OR 1.23% APF	38% SDF
Root permanent	5000 ppm/1.1% NaF toothpaste or gel; SDF	5000 ppm/1.1% NaF toothpaste or gel; SDF

Mechanism Pathways
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Caries Development
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- Biofilm bacteria colonize tooth surface.
- Fermentable carbohydrate becomes available.
- Bacteria produce organic acids.
- Plaque pH falls.
- If pH drops below critical threshold repeatedly, enamel minerals dissolve.
- Saliva/fluoride/calcium/phosphate can remineralize early damage.
- Continued imbalance produces visible white spot, staining, surface breakdown, cavitation, and possible pulpal involvement.

Stephan Curve Logic
 [Back to TOC](#)

VISUAL PATHWAY: Core Framework
carbohydrate exposure v plaque pH drops quickly v

```
if pH < 5.5 -> enamel demineralization favored
    |
    v
saliva buffering + clearance slowly raises pH
    |
    v
repeated frequent exposures = less recovery time = higher caries risk
```

Visual Diagnosis Pathway [Back to TOC](#)

1. Clean and dry.
2. Inspect systematically.
3. Identify location: occlusal, proximal, smooth, cervical, root, restoration margin.
4. Identify stage: white spot, stained, cavitated, arrested.
5. Identify activity and surface integrity.
6. Use radiographs as needed.
7. Use risk level to decide observation/prevention/non-restorative/restorative path.

Pulpal Disease Progression [Back to TOC](#)

1. Caries deepens.
2. Bacterial by-products diffuse through dentinal tubules.
3. Chronic inflammatory infiltrate develops.
4. Vasodilation/permeability increases.
5. Tissue pressure rises in rigid pulp chamber.
6. Vessel compression worsens circulation.
7. Entire pulp can become involved.
8. Necrosis and periapical spread may follow.

CRA Management Sequence [Back to TOC](#)

1. Risk factors + protective factors are gathered.
2. Disease indicators are identified.
3. Risk level assigned.
4. Recall/bitewing schedule chosen.
5. Prevention/products matched to risk.
6. Patient-specific behavior changes are planned.
7. Reassess risk at recall.

Risk-Based Product Escalation [Back to TOC](#)

VISUAL PATHWAY: Core Framework

```
LOW
  nutrition counseling + OTC fluoride toothpaste 2x/day

MODERATE
  low regimen + OTC 0.05% NaF rinse + 5% NaF varnish
```

HIGH
moderate regimen + prescription 1.1% NaF / 5000 ppm toothpaste

EXTREME
high regimen + dry-mouth rinse/buffers + salivary/bacterial adjuncts as indicated

Professor Emphasis [Back to TOC](#)

PROFESSOR SIGNAL

Use this section as the priority map. If time is tight, drill these topics first.

Topic	What to focus on	Why it matters	Exam move
Caries sequence	Bacteria ferment carbohydrate -> acid -> demineralization.	Repeated as the main mechanistic core.	Write the sequence, not just "sugar causes cavities."
Critical pH	Enamel begins dissolving around pH 5.5 .	Explicitly highlighted as important.	Memorize 5.5 .
White spot lesions	Early lesions can be reversed/arrested.	Called a major take-home concept.	Non-cavitated/intact = prevention/remineralization.
Diagnosis	Visual exam first; tests are imperfect.	Professor discussed sensitivity/specificity and test limits.	Use combined diagnosis, not one test.
DMFT/DMFS	Severity/intensity, not prevalence.	Repeated professor correction.	If asked prevalence, do not answer DMFT.
Untreated caries	Focus more on untreated disease patterns than raw prevalence numbers.	Professor repeatedly de-emphasized raw prevalence numbers.	Know subgroup trends, not every percentage.
Water fluoridation	Definition is testable.	Professor emphasis explicitly says definition will be on exam.	Controlled addition to public water supply to prevent caries while avoiding fluorosis .
Fluoride products	Know product choices by risk level.	Professor emphasis says low vs high risk products matter.	Match risk level to OTC/rinse/varnish/Rx fluoride.
SDF	Know side effects/disadvantages before recommending.	Professor says patient explanation is required.	Black staining, silver allergy, mucosal ulceration/stomatitis, mild gastric effects.
Diet	Sticky between-meal sugar and fermentable carbohydrate drive risk.	Diet examples and studies emphasized.	Frequency/stickiness > sugar yes/no.
Saliva	Salivary functions will be tested.	Professor emphasis says questions will cover salivary functions.	Clearance, buffering, minerals, antimicrobial proteins, remineralization.
Sjogren /medications	Know dry-mouth causes and implications.	Professor emphasis calls Sjogren important and	Connect dry mouth to high risk and

Topic	What to focus on	Why it matters	Exam move
		asks for medication categories.	flow testing.
CRA	Risk is balance of protective and risk factors; not one test.	Repeated warning.	Define CRA by balance; use history/exam/radiographs/form/tests.
CRA time frame	Last 36 months / 3 years matters.	Professor told students to remember it.	Apply to new lesions/restorations/missing teeth due to caries.
Recall intervals	Low/moderate/high/extreme recall and bitewings.	Professor emphasis says these numbers need to be known.	Memorize 12/24, 6/18, 4/12, 3/6 .
Resin infiltration	Incipient/white lesions; smooth/approximal surfaces.	Professor emphasis flags use and surface specificity.	Not for cavitated bulk restoration.

Last-Pass Checklist [Back to TOC](#)

Definitions [Back to TOC](#)

- ☐ Dental caries.
- ☐ Demineralization and remineralization.
- ☐ Stephan curve.
- ☐ **CAMBRA** and CRA.
- ☐ Prevalence, incidence, **DMFT/DMFS**.
- ☐ Risk factor and protective factor.
- ☐ Fluorosis and water fluoridation.
- ☐ Xerostomia and **Sjogren** syndrome.

Mechanisms [Back to TOC](#)

- ☐ Biofilm + carbohydrate + time -> acid -> demineralization.
- ☐ Critical pH **5.5**.
- ☐ Saliva buffering and mineral support.
- ☐ Fluoride remineralization/reduced demineralization.
- ☐ Dentinal tubules and pulp irritation.
- ☐ Pulp pressure -> vessel compression -> necrosis.

Clinical Recognition [Back to TOC](#)

- ☐ White spot lesion.
- ☐ Arrested lesion.
- ☐ Root caries.
- ☐ Recurrent caries.
- ☐ Lower first molar susceptibility.
- ☐ Lower canine relative protection.
- ☐ Reversible vs irreversible pulpitis vs necrosis.

Comparisons [Back to TOC](#)

- ☐ Active vs arrested.
- ☐ Initial vs cavitated.
- ☐ Total vs untreated caries.
- ☐ Prevalence vs DMFT.
- ☐ Systemic vs topical fluoride.
- ☐ NaF vs APF vs SnF2 vs **SDF**.
- ☐ A-delta vs C fibers.
- ☐ Diet vs nutrition.

Risk Assessment and Management [Back to TOC](#)

- ☐ Risk is not determined by one test.
- ☐ Use **36-month** window.
- ☐ Know low/moderate/high/extreme risk levels.
- ☐ Know recall/bitewing intervals.
- ☐ Know dry-mouth threshold **<0.7 ml/min** by saliva flow test.
- ☐ Match product plan to risk level.

Fluoride and Prevention [Back to TOC](#)

- ☐ Water fluoridation definition and **0.7 ppm**.
- ☐ Toothpaste/rinse/gel/varnish/**SDF** roles.
- ☐ 5% NaF varnish.
- ☐ **38% SDF**.
- ☐ **5000 ppm** prescription fluoride.
- ☐ **SDF** side effects.
- ☐ Chlorhexidine side effects.

Saliva and Diet [Back to TOC](#)

- ☐ Normal flow rates.
- ☐ Saliva protective functions.
- ☐ IgA inhibits adhesion.
- ☐ Medication categories causing low saliva.
- ☐ Vipeholm conclusions.
- ☐ Xylitol actions and limits.